

Chemistry of Lipid Nanoparticles for RNA Delivery

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Yulia Eygeris,^{||} Mohit Gupta,^{||} Jeonghwan Kim, and Gaurav Sahay*



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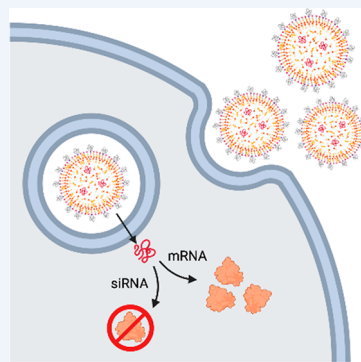
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CONSPECTUS: Lipid nanoparticles (LNPs) are a type of lipid vesicles that possess a homogeneous lipid core. These vesicles are widely used in small-molecule drug and nucleic acid delivery and recently gained much attention because of their remarkable success as a delivery platform for COVID-19 mRNA vaccines. Nonetheless, the utility of transient protein expression induced by mRNA extends far beyond vaccines against infectious diseases—they also hold promise as cancer vaccines, protein replacement therapies, and gene editing components for rare genetic diseases. However, naked mRNA is inherently unstable and prone to rapid degradation by nucleases and self-hydrolysis. Encapsulation of mRNA within LNPs protects mRNA from extracellular ribonucleases and assists with intracellular mRNA delivery. In this Account, we discuss the core features of LNPs for RNA delivery. We focus our attention on LNPs designed to deliver mRNA; however, we also include examples of siRNA-LNP delivery where appropriate to highlight the commonalities and the dissimilarities due to the nucleic acid structure. First, we introduce the concept of LNPs, the advantages and disadvantages of utilizing nucleic acids as therapeutic agents, and the general reasoning behind the molecular makeup of LNPs. We also briefly highlight the most recent clinical successes of LNP-based nucleic acid therapies. Second, we describe the theory and methods of LNP self-assembly. The common idea behind all of the preparation methods is inducing electrostatic interactions between the nucleic acid and charged lipids and promoting nanoparticle growth via hydrophobic interactions. Third, we break down the LNP composition with special attention to the fundamental properties and purposes of each component. This includes the identified molecular design criteria, commercial sourcing, impact on intracellular trafficking, and contribution to the properties of LNPs. One of the key components of LNPs is ionizable lipids, which initiate electrostatic binding with endosomal membranes and facilitate cytosolic release; however, the roles of other lipid components should not be disregarded, as they are associated with stability, clearance, and distribution of LNPs. Fourth, we review the attributes of LNP constructs as a whole that can heavily influence RNA delivery. These attributes are LNP size, charge, internal structure, lipid packing, lipid membrane hydration, stability, and affinity toward biomacromolecules. We also discuss the specific techniques used to examine these attributes and how they can be adjusted. Finally, we offer our perspective on the future of RNA therapies and some questions that remain in the realm of LNP formulation and optimization.



KEY REFERENCES

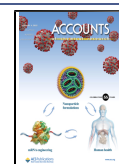
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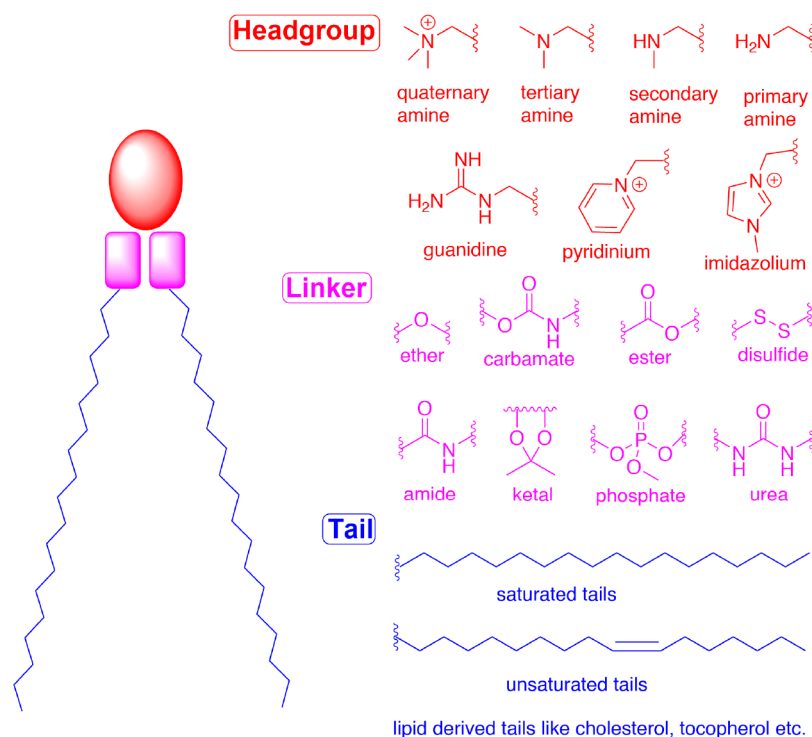


Figure 1. Schematic representation of CLs and ILs and their components (headgroup, linker, and tails).

analogues using a Gal8-GFP assay. LNPs with a polymorphic shape or multilamellar organization caused a greater extent of endosomal disruption as indicated by galectin recruitment.

- Kim, J.; Jozic, A.; Sahay, G. Naturally Derived Membrane Lipids Impact Nanoparticle-Based Messenger RNA Delivery. *Cell. Mol. Bioeng.* **2020**, 13 (5), 463–474.⁴ Continuing the investigation of the structural lipids, this study examines the effect of naturally occurring membrane lipids in LNP formulations. These lipids demonstrated disparities depending on the mode of administration.

INTRODUCTION

In recent years, lipid nanoparticles (LNPs) have demonstrated their utility as a delivery platform for RNA vaccines and therapies.⁵ Naked RNA is a negatively charged hydrophilic macromolecule that struggles to enter cells because of the electrostatic repulsion by the cellular membrane and is rapidly degraded by omnipresent RNases. Therefore, it requires a protective shell to gain access to the cell interior. Since cellular membranes are primarily composed of lipids, encapsulating RNA with lipid vesicles offers the opportunity to covertly pass the cellular membrane and released RNA to the cytosol. To do so, the vesicles first need a positively charged lipid that would latch onto the negatively charged RNA. However, the vesicles consisting of permanently cationic lipids can cause cytotoxicity due to electrostatic disruption of the negatively charged cellular membrane. The lipid structures then have evolved further to acquire positive charge in response to the acidic endolysosomal pathways. LNP composition has also expanded to include structural lipids (to mimic the cell membrane and shield the positive charge) and poly(ethylene glycol)-anchored lipids (to prevent the LNPs from aggregation and undesired

interactions with biological environments). As is evident from the major regulatory milestones in the field—namely, FDA approval of the first siRNA-LNP drug (Onpattro) and mRNA-LNP COVID-19 vaccine (Comirnaty) and emergency use authorization (EUA) of Moderna's COVID-19 vaccine—the LNP-based approach to nucleic acid delivery is safe and amenable to various therapeutic cargos. Nonetheless, there is currently no one-fit-all solution to suit every disease, so continuous work on LNP optimization is ongoing. This Account will discuss the major scientific and manufacturing concepts needed to select lipids for RNA-LNP delivery.

PREPARATION

LNP preparation relies on the power of self-assembly, that is, the spontaneous organization of lipid components into nanostructured entities based on intermolecular interactions.⁶ LNP formation begins with the electrostatic association between the negatively charged nucleic acids and positively charged lipids. Then, LNPs grow via hydrophobic and van der Waals interactions between lipid components. Characterization of the early stages of self-assembly and the relevant effects on the final LNP properties remains challenging because of the diversity of the lipid chemistry, the unique properties of nucleic acids, and the time scale at which the mixing of the two occurs. In addition, LNP manufacturing protocols affect the products of self-assembly in at least two ways: the homogeneity of the LNPs and the nucleic acid loading efficiency.

LNP preparation can be done in many ways, such as extrusion of lipid vesicles, rehydration of the lipid film, nanoprecipitation, and microfluidic mixing. However, the general preparation is some form of rapid mixing of aqueous and lipid components.⁷ Microfluidic methods are preferred nowadays for preclinical studies because of their good reproducibility. Recent advances in the manufacturing of microfluidic devices have made this method more accessible.^{8,9}

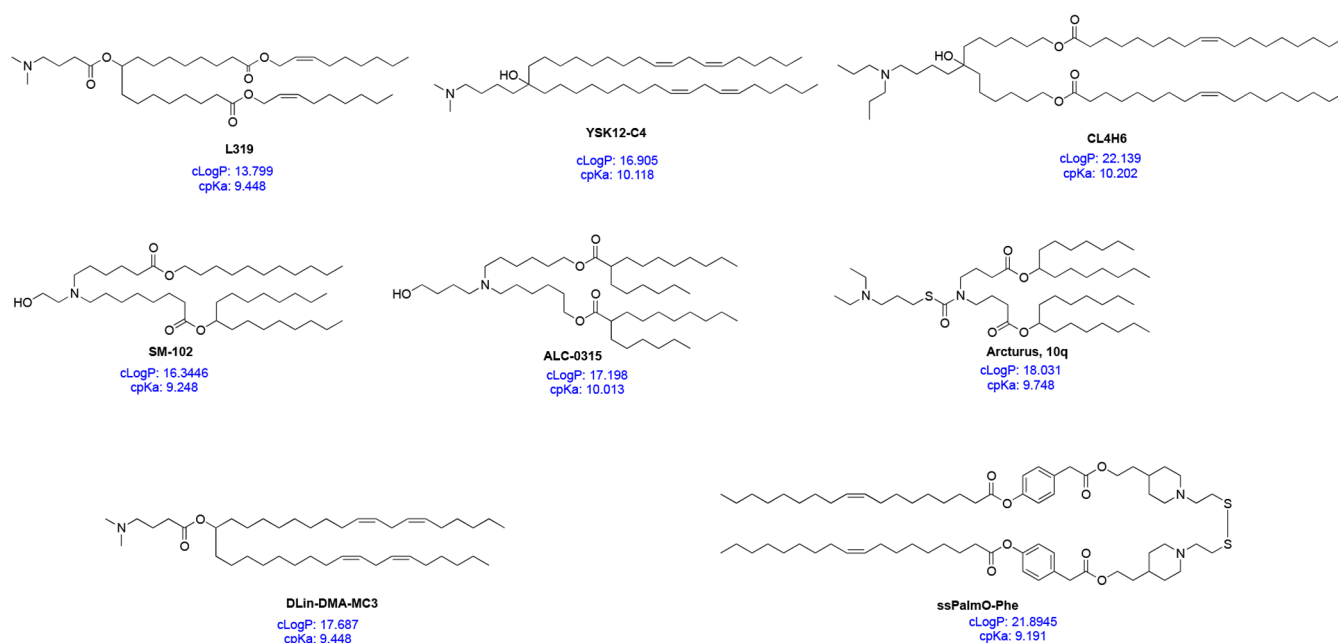


Figure 2. Structures and cpKa and cLogP values for selected ILs.

Employing parallel microfluidic paths¹⁰ or revised conventional approaches like pipet mixing¹¹ and T-mixers are other approaches to achieve even more high-throughput preparation of LNPs. While microfluidic mixing is generally beneficial to encapsulate hydrophilic moieties into the hydrophobic lipid core, it is not strictly required to incorporate nucleic acids. Kulkarni et al. reported that LNPs could incorporate siRNA via a T-mixer once the lipid core is formed.¹² Overall, the nature of the mixing method may influence the loading efficiency and the internal organization of LNPs, as the nanoparticle structure appears to be determined by kinetic factors of self-assembly.¹³

■ FORMULATION COMPONENTS

Cationic and Ionizable Lipids

Cationic lipids (CLs) and ionizable lipids (ILs) initiate the first step of self-assembly via electrostatic interactions. Lipoplexes containing CLs are still widely used to deliver nucleic acids.¹⁴ However, because of toxicity issues and lack of in vivo efficacy, they have been replaced by pH-sensitive ILs. When formulated into LNPs, these lipids are designed to show net neutrality at physiological pH but become positively charged inside the acidic endosomes. The pH-dependent ionizability makes them suitable materials for nucleic acid delivery because of the improved efficacy and toxicity profile.^{14–16} These lipids typically constitute 30–50% of the total lipids in a formulation.^{17–20} Much research has been devoted to fine-tuning the properties of the ILs to improve the efficiency further, especially in hard-to-reach tissues.¹⁷ The overall architecture of CLs and ILs can be broken down into three parts: (1) the headgroup, (2) the linker, and (3) the tails (Figure 1).

Headgroups. The headgroups of ILs typically have a positive charge. The dimensions and charge density of the headgroup are responsible for entrapping the nucleic acid, stabilizing the LNP, interacting with the cell membrane, and facilitating the endosomal escape.²¹ ILs may contain several ionizable headgroups, although ILs with one headgroup are

more common. The typical headgroups include amines (from primary to quaternary), guanidine, and heterocyclic groups^{17,21} (see Figure 1). ILs in clinics (DLin-MC3-DMA,²² SM-102,²³ and ALC-0315;²⁴ see Figure 2) contain a tertiary amine headgroup, which shows pH-dependent ionization. The headgroups of ALC-0315 and SM-102 also contain a terminal hydroxyl group which could decrease the hydration of the headgroup and improve hydrogen-bonding interactions with the nucleic acid, potentially leading to improved transfection ability.^{25,26}

Linkers. The linker typically connects the headgroup with the tails, although the linkers may also be buried within the tails (SM-102 and ALC-0315; Figure 2). The linker impacts the stability, biodegradability, cytotoxicity, and transfection efficiency of LNPs.^{17,21} Common linkers utilized in the design of CLs and ILs are shown in Figure 1. ILs may contain one or several linkers; however, most ILs contain only one type of linker, which may be due to ease of synthesis. The linkers can be classified as non-biodegradable (e.g., ethers and carbamates) and biodegradable (e.g., esters, amides, and thiols). Biodegradable linkers are preferred because they usually show rapid clearance in vivo, enabling multiple dosing and reducing the propensity for side effects.^{6,27} Notably, DLin-MC3-DMA, ALC-0315, and SM-102 all have ester linkers. For SM-102, modifications around the ester groups were shown to affect the clearance, formulation stability, and transfection efficiency of the LNPs.²³

Tails. The hydrophobic tails impact the pK_a, lipophilicity, fluidity, and fusogenicity, thereby influencing the nanoparticle formation and potency.^{17,21} Typically, an IL incorporates one to four hydrophobic tails containing 8 to 20 carbon atoms.^{17,21} The tails may be saturated or unsaturated. The degree of unsaturation has been shown to influence nucleic acid delivery by modulating interrelated aspects of membrane destabilization.^{17,28} DLin-MC3-DMA has two linoleyl tails, while ALC-0315 and SM-102 contain two branched saturated tails that are postulated to endow a cone-shaped geometry, facilitating

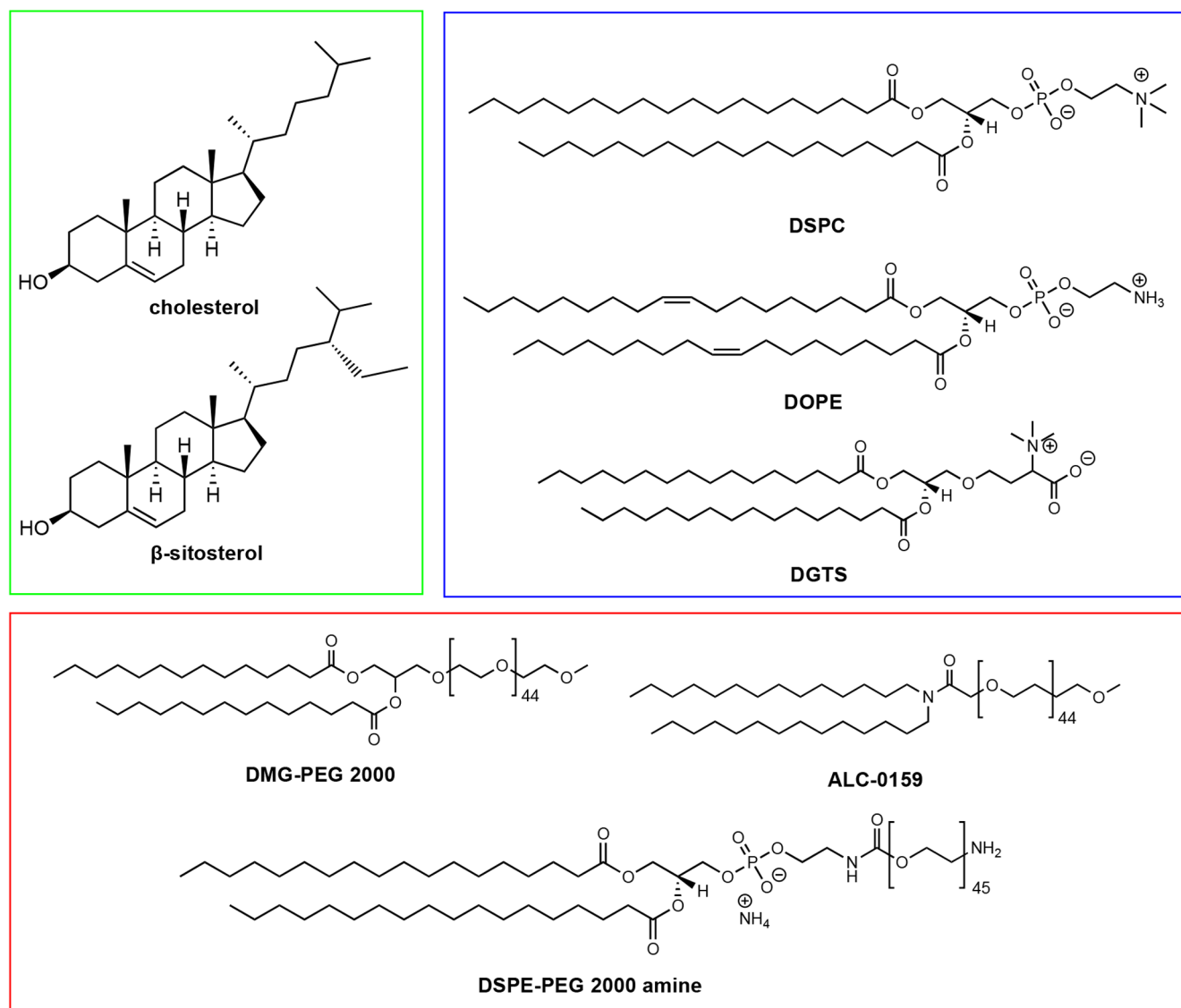


Figure 3. Examples of sterols (green), phospholipids (blue), and PEG-lipids (red) used in LNP formulations.

destabilization of the endosomal membrane and cytosolic release of the nucleic acid.²⁷

ILs may be considered as multicomponent molecules in which each part has to be precisely designed to package and deliver the nucleic acid safely and efficiently. Understanding the properties of the ILs as a whole may also aid in the design of next-generation ILs. One of these properties is the calculated pK_a (cpKa) of the ILs, which can be readily determined in silico. The cpKa values for common ILs range from 9 to 10.5 (Figure 2). A recent study showed that the cpKa closely matches the actual pK_a of the IL.^{27,29} It appears that the cpKa of the ILs influences the overall pK_a of the corresponding LNP formulation, such that when the cpKa of the IL is about 8.5–10.5, the LNP shows a pK_a value of approximately 6–7.³⁰ The difference between the IL cpKa and LNP pK_a appears to be consistent and equal to 2–4 units.^{27,29} Thus, cpKa may be used as a guiding tool for the design of new ILs. Two other less appreciated properties of the ILs are the cLogP and cLogD values, which represent the lipophilicity of a molecule in the un-ionized and ionized states, respectively. In one recent study, Rajappan et al. investigated the effects of pK_a ,

cpKa, and cLogD on siRNA delivery by LNPs and found that the most potent lipids have cLogD in the range of 10–14.³¹ Since some common ILs have cLogP values in the range of 15–20 (Figure 2), the lipophilicity of the ILs should also be considered for the design of the next ILs. Since the ionizability (cpKa) and lipophilicity (cLogP) of the IL can impact processes beginning from the formation of the initial complex with the nucleic acid to the final nanoparticle formation and the delivery of the cargo, considering both of these values simultaneously prior to the IL synthesis may speed up the discovery of a potent IL. Nevertheless, more studies are needed in order to cement this observation.

In addition to the traditional CLs and ILs, there are also examples of zwitterionic lipids.¹⁷ In a recent study, Liu et al. synthesized a library of more than 500 zwitterionic lipids called iPhos.³² These iPhos are composed of an amine group, short hydrophobic tails, and a phosphate linker. It is proposed that the negatively charged phosphate group facilitates membrane fusion and leads to endosomal escape. Various formulations of the top candidate, 9A1P9, could preferentially deliver the nucleic acid of interest to the liver and lungs.

In summary, the properties of the individual parts of the ILs influence the overall formulation and biological characteristics. Numerous systemic studies have been carried out over the last 50 years to design the ideal IL. Some of these ILs have been approved by the FDA for the delivery of genetic cargo.^{15,17} However, much research is still needed to design ILs capable of delivering different types of genetic cargo with higher efficiency in nonhepatic targets without toxicity. Readers seeking more information about the synthesis of ILs may direct their attention to an excellent review on the matter.¹⁷

Sterols

Cholesterol, a naturally abundant component of cellular membranes, is one of the so-called structural lipids that is often used in LNP formulations. Cholesterol typically makes up about 20–50% of the total lipids in an LNP formulation.^{17–20} An abundant component of animal cellular membranes, cholesterol is commonly extracted in bulk from natural sources such as sheep wool. Despite its natural abundance, the role of cholesterol in cellular uptake has been overlooked until recently. Interestingly, replacing cholesterol with naturally occurring phytosterols such as β -sitosterol¹ (see Figure 3) and oxidized cholesterol derivatives³³ yielded significant improvements in mRNA delivery, likely determined by differential lipid trafficking in the endocytic pathway^{34,35} and enhanced endosomal escape of the LNPs.³ While it is not entirely clear why LNPs containing β -sitosterol are more likely to escape the endolysosomal pathway, their polymorphic shape and multilamellar organization^{1,2} may deform the endosomal membrane or extend the time frame of nucleic acid release. Cholesterol predominantly resides in the outer shell of the LNP,³⁶ which explains why the modification to the sterol structure may induce the changes in the LNP organization at the surface. Moreover, a recent study suggested that cholesterol migrates to the outer lipid shell from the LNP core upon exposure to apolipoprotein E (ApoE).³⁶ These observations suggest that cholesterol and its derivatives may influence cellular recognition pathways despite cholesterol's relative inertness compared to ionizable lipids. However, many unknowns and hurdles remain with regard to the potential modifications of cholesterol in LNP formulations. For example, phytosterols are not as abundant as cholesterol, and their sparse supply and high cost in production may hinder their utilization for LNP formulations. Similarly, the nature of crude materials and consequential trace impurities may contribute to the potential batch-to-batch variability of LNPs. The stereochemistry of sterols and lipids in general is a fascinating aspect to explore in LNP formulations. Does stereochemistry affect lipid recognition, trafficking, and recycling? Do enantiomers have significant differences in the lipid packing at the LNP surface? We hope to learn the answers to these questions soon.

Phospholipids

Phospholipids help package nucleic acids and stabilize LNPs. They are relatively less explored than other lipid components and typically constitute only about 10–20% of the total lipids in a formulation.^{17–20} Phospholipids are used as structural lipids because of their spontaneous organization into lipid bilayers and high phase transition temperatures that confer the membrane stability of LNPs. As with cellular membranes, phospholipids are located at the periphery of LNPs. These lipids are often semisynthetic. For example, phosphatidylcholine is typically sourced from natural sources such as egg yolk and soybeans and can be chemically modified to include fatty

acid tails. 1,2-Distearoyl-*sn*-glycero-3-phosphocholine (DSPC) (see Figure 3) is a structural lipid that is used in clinically approved LNPs, such as a siRNA therapeutic (Onpatro) and mRNA vaccines against SARS-CoV-2.⁶ DSPC is structurally composed of a phosphatidylcholine headgroup and two saturated 18-carbon tails that form a tightly stacked lipid bilayer. In LNPs, it is located mainly on the nanoparticle surface and marginally in the nanoparticle core.^{37,38} 1,2-Dioleoyl-*sn*-glycero-3-phosphoethanolamine (DOPE) is another phospholipid frequently used in preclinical studies of LNPs. DOPE not only forms a more fluid lipid layer because of the unsaturated tails but also has the intrinsic ability to organize the hexagonal II (H_{II}) phase. The H_{II} phase is thought to facilitate the membrane fusion of the lipid membrane to the endosomal membrane, resulting in the cytosolic release of the nucleic acid cargo. Several studies showed that DOPE improves the RNA transfection efficiency compared with DSPC in lipidoid-based LNPs.^{39,40} Recently, Zhang et al. reported that DOPE led to the hepatic accumulation of C12-200 LNPs, whereas DSPC led to splenic accumulation upon intravenous administration, suggesting the influence of the structural lipid on LNP biodistribution.²⁰ We also found that the replacement of DSPC in MC3-based LNPs with naturally occurring glycolipids influences mRNA transfection, and 1,2-dipalmitoyl-*sn*-glycero-3-*O*-4'-(*N,N,N*-trimethyl)homoserine (DGTS) (see Figure 3), a plant-derived membrane lipid, shows different transfection efficiencies depending on the administration route.⁴ Overall, these studies underline the significance of structural lipids in LNP-mediated RNA delivery.

PEG-Anchored Lipids

An important module of LNPs that controls their half-life and cellular uptake is PEG-anchored lipids (PEG-lipids; see the examples in Figure 3). During LNP assembly, the PEG chain is located in the outer shell of the nanoparticle because of its hydrophilicity and bulkiness. As with other nanocarriers, PEG provides LNPs with an exterior polymeric layer to hinder adsorption of serum proteins and the mononuclear phagocyte system, extending the circulation time in vivo. PEG also prevents aggregation of the nanoparticles in storage and in the bloodstream. Additionally, the amount of PEG-lipids may determine the particle size. Another potential purpose of PEG-lipids is to functionalize the surface of LNPs. Functionalized PEG-lipids enable bioconjugation of LNPs with ligands or biomacromolecules. For example, Singh et al. used DSPE-PEG-amine to conjugate hyaluronan via NHS/EDC chemistry for tumor targeting,⁴¹ and Parhiz et al. used DSPE-PEG-maleimide to conjugate antibodies via SATA-maleimide chemistry.⁴² While PEG is useful for LNP stability and bioconjugation, its desorption is also critical for cell transfection. Shedding of PEG from LNPs allows opsonization by serum proteins, such as apolipoproteins and albumins, which are the key effectors of receptor-mediated endocytosis of LNPs.^{43,44} Akinc et al. demonstrated that ApoE binds to the LNPs, leading to low-density lipoprotein receptor (LDLR)-mediated internalization to hepatocytes.⁴⁴ Since PEG-lipids inhibit binding of ApoE to LNPs, having an excessive amount of PEG-lipids can adversely impact cellular uptake and transfection of LNPs. The LNPs containing fewer PEG-lipids appear to be more efficient in delivering nucleic acid because of facile binding to ApoE.^{45–47} The length of lipid anchors of PEG-lipids is also an important factor determining the desorption rate. Mui et al. reported that shedding of PEG

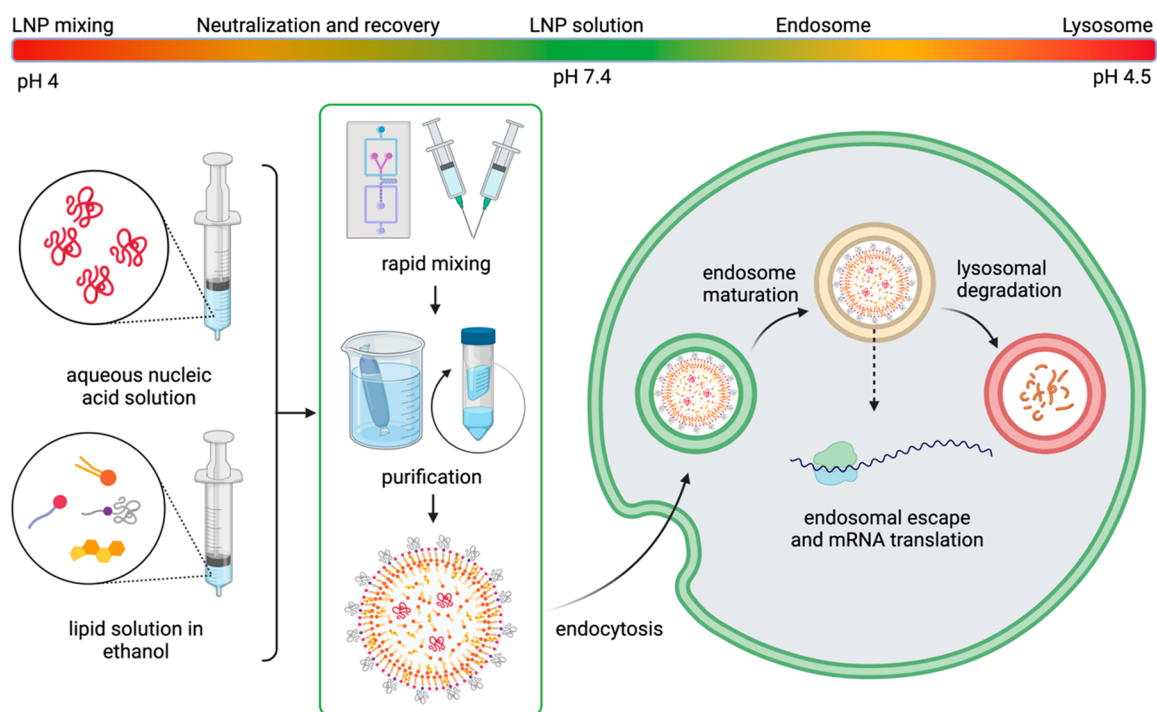


Figure 4. pH fluctuations that an LNP encounters in its life cycle. Figure created with BioRender.

from the LNP membrane is inversely proportional to the length of the lipid anchor in PEG-lipids because the hydrophobic interactions between the PEG-lipid and LNP membrane increase with increasing anchor length of the PEG lipid.⁴⁸ The rate of the PEG shedding may also influence the production of anti-PEG antibodies and pose complications for repeated administration, as suggested by Suzuki et al.⁴⁹ It is rare to see LNP formulations for intravenous administration contain more than 2% PEG-lipids; however, the dense PEG layer could be beneficial to reach extrahepatic targets. Lee et al. showed that tumor accumulation of LNPs was higher at 5% PEG-lipid than at 2.5%,⁵⁰ and Lokugamage et al. demonstrated the critical importance of PEG-lipids for delivery of nebulized LNPs.⁵¹ Consequently, the amount and type of PEG-lipid in LNPs may require careful adjustment based on the clinical needs.

■ PROPERTIES

The mean size and size distribution of LNPs are important initial determinants of the LNP quality and suitability for various applications. These properties are usually investigated by dynamic light scattering (DLS). Generally, the optimal LNP size is 20–200 nm, as this size makes them sufficiently robust to withstand the fluid flow (e.g., blood and lymph) while allowing LNPs to cross the interstitium.^{52–54} The LNP size is often modulated by changing the quantity of PEG-lipids or the mixing parameters (e.g., flow rate and volume ratio). The size may affect the internalization, biodistribution, degradation, and clearance of LNPs, and diverse particle sizes may be required for different applications. For example, 45 nm siRNA LNPs were the most potent for subcutaneous delivery, while 80 nm siRNA LNPs were the most potent for intravenous delivery in mice.^{46,55} However, a comparison of various mRNA-LNP particle sizes in rodents and nonhuman primates revealed that nonhuman primates were less sensitive to the particle size when LNPs were delivered intramuscularly.⁵⁶

LNP surface charge is responsible for interactions with cellular membranes and the biological environment. Since the cell membrane is negatively charged, LNPs with negative surface charge are repelled from the membrane and do not get taken up by the cells. On the other hand, positively charged LNPs may directly disrupt cellular membranes and cause cytotoxicity. This is why ionizable lipids are crucial in the LNP design: initially, LNPs containing ionizable lipids are neutral and avoid any unwanted electrostatic interactions but acquire a positive charge at acidic endosomal pH. The surface charge of LNPs is commonly evaluated by zeta potential measurements. This technique is typically used to evaluate colloidal aggregation, and although there is no strict classification, the surface charge is considered weak if the zeta potential falls between -20 and $+20$ mV. A common way to adjust the total surface charge of LNPs is to adjust the N/P ratio or the ratio of ionizable lipid (N, for cationic amines) to nucleic acid (P, for anionic phosphates). Carrasco et al. reported that increasing the N/P ratio in LNPs containing the ionizable lipid KC2 increased the surface charge and encapsulation efficiency.²⁹ Interestingly, incorporating permanently charged lipids into the LNPs may change preferential organ uptake without adding a surface charge. Cheng et al. achieved selective organ targeting (SORT) in mice based on the lipid charge: addition of positively charged lipids to LNP formulations led to the preferential transfection of lung tissues, while negatively charged lipids directed transfection toward the spleen.¹⁸

Lipid packing may impact a myriad of parameters, from membrane hydration and deformability to cellular uptake and cargo release. The basics of lipid organization required to form a lipid vesicle were summarized in a recent review.⁵⁷ Briefly, every lipid can be described by a packing parameter that depends on the volumes occupied by the lipid polar “head” and nonpolar “tail”. Structurally equalized lipids form cylindrical structures and lamellar phases, while “unbalanced” structures induce hexagonal, cubic, and micellar phases. The inverted

hexagonal phase (H_{II}) appears to most significantly enhance lipid membrane fusion.^{58,59} To date, controlled fabrication of nonlamellar phases remains relatively unusual in the realm of RNA-LNP delivery, with cubosomes being the most prominent example.^{60,61} However, LNPs may undergo structural changes upon exposure to environmental triggers. Heyes et al. investigated the phase transition behavior of lipid particles containing a series of cationic lipids with various lipid tails using ^{31}P NMR spectroscopy and found that lipids with lower lamellar-to-inverted-hexagonal transition temperatures (T_{BH}) were more fusogenic, as evidenced by the gene silencing efficiency.²⁸ Similarly, one of the theories of membrane destabilization proposes that as ionizable lipids are exposed to the acidic pH of late endosomes, electrostatic interactions between the ionizable lipids and phospholipids in the endosomal membrane cause membrane breach.⁶² A recent work by Liu and colleagues utilized this concept and reported evidence for the formation of the inverted hexagonal phase as indicated by ^{31}P NMR spectroscopy when one of the novel ionizable lipids was exposed to the endosomal mimic.³² While pH-induced association is the most common mechanism reported for lipid materials, additional approaches to endosomal membrane destabilization are discussed in our recent review.⁶³

Depending on the lipid phase and overall polarity, the lipid membrane may entrap water and alter the membrane fluidity or deformability, which may affect the lipid membrane fusion.⁶⁴ Membrane hydration may also affect the potential responsiveness to pH, a critical environmental trigger that is commonly exploited to initiate nucleic acid release. The pH fluctuations that an LNP encounters in its lifetime are illustrated in Figure 4. As LNPs enter the intracellular space, they are entrapped inside the endosomes, which gradually acidify as they mature into lysosomes. Higher water content inside the lipid membrane thus may affect the kinetics of acidification and assist rapid membrane destabilization. Koitabashi et al. examined lipid membrane destabilization in response to pH in siRNA-LNPs using a Laurdan assay and found a positive correlation between membrane hydration and gene silencing efficiency;⁶⁵ however, this report did not focus on the kinetics of acidification. An interesting point regarding membrane hydration is that siRNA-LNPs have less water content than mRNA-LNPs of identical formulation as suggested by ^{31}P NMR experiments,⁶⁶ which likely stems from the much longer length of hydrophilic RNA chains. These observations were further corroborated by Carrasco et al., who observed that KC2 LNPs with a low N/P ratio contained a higher number of mRNA and lipids per nanoparticle and had a higher dielectric constant, suggesting that LNPs with a low N/P ratio are more hydrated than LNPs with a high N/P ratio.²⁹ Higher RNA solvation also led to an improvement in transfection.²⁹ Thus, mRNA-LNPs may be more sensitive to environmental changes, although the changes in pH sensitivity may be irrelevant on the time scale of biological processes. The reorganization upon exposure to the biological environment further complicates the question of LNP shell hydration.³⁶

The inherent aqueous environment of LNPs also poses a threat to their long-term stability. Pure nucleic acids may quickly deteriorate under ambient conditions via exogenous RNase degradation or self-hydrolysis. Although LNPs can protect nucleic acids from enzymatic degradation, they are prone to aggregation due to thermodynamic factors (such as

minimizing phase separation), which may lead to loss of the nucleic acid upon nanoparticle fusion and ultimately compromise the transfection efficiency.^{67,68} Low-temperature storage and lyophilization may preserve RNA but damage LNPs as a result of ice crystallization, although incorporating a cryoprotectant such as sucrose seems to alleviate this problem.^{67,68} Interestingly, LNPs may change their preferential organ uptake depending on the storage conditions,⁶⁸ perhaps as a result of their reorganization. From a practical perspective, recently developed mRNA-LNP vaccines against COVID-19 created a logistics infrastructure for RNA therapeutics, which may eliminate many concerns regarding LNP stability, storage, and transportation. These vaccines promise long-term stability for up to 6 months at freezer temperature ($-20\text{ }^{\circ}\text{C}$) and up to 30 days at room temperature, dramatically improving access to these groundbreaking treatments. Notably, the Pfizer/BioNTech and Moderna vaccines have different storage requirements, suggesting that variations in LNP formulation may significantly alter the nucleic acid affinity and LNP stability. While there is currently no established methodology for accelerated stability testing of these amorphous materials—meaning, the stability has to be evaluated empirically at discrete time points—thermal methods such as differential scanning calorimetry (DSC) may provide valuable insights into LNP degradation. Ultimately, a better understanding of LNP self-assembly is required to project the downstream properties of LNP-based RNA therapeutics.

The internal organization of lipid nanoparticles is a matter that to date has not been universally resolved. Several reports indicate that siRNA-LNPs have a “sandwiched” structure in which siRNA is associated with lipid layers at the exterior of the LNPs.^{37,69} On the other hand, mRNA may also stretch along the LNP perimeter, occupy the water–lipid interface inside “blebbed” LNPs,⁷⁰ or occupy interconnected aqueous channels,^{38,66} where mRNA could get displaced due to its hydrophilicity. mRNA also may get alienated from the bulk of the lipids into aqueous pockets.^{2,70} Our recent study utilizing cryo-TEM suggests that any deviation from a homogeneous, spherical LNP (e.g., polymorphic shape, multilamellar structure, or aqueous pockets) may improve the mRNA transfection efficiency;² however, it is not apparent whether the improved transfection may have been affected by the molecular recognition of lipid components. The changes in the lipid organization may also be affected by solubility. Although the lipids are typically solubilized before mixing either by heat or sonication, Yanez Arteta et al. reported the presence of cholesterol 2D crystals in the LNP lipid membrane as detected by small-angle X-ray scattering.³⁸ Lipid organization may also change upon exposure to the biological environment, i.e., LNPs rearranged both at the surface and in the core upon exposure to ApoE, an abundant protein in blood serum, as detected by small-angle neutron scattering experiments.³⁶ The uncertainty surrounding the question of lipid organization upon RNA delivery calls for further investigations.

Nanoparticles are prone to nonspecific adsorption of proteins, resulting in the formation of the biomolecular corona at the interface. The formation of the biomolecular corona on nanoparticles changes the surface properties and physicochemical characteristics of the nanoparticles and plays a significant role in biodistribution and endocytosis. ApoE is one component of the biomolecular corona that is well-known to affect LNP-mediated nucleic acid delivery. ApoE is most abundant in the liver, synthesized mainly through hepatocytes,

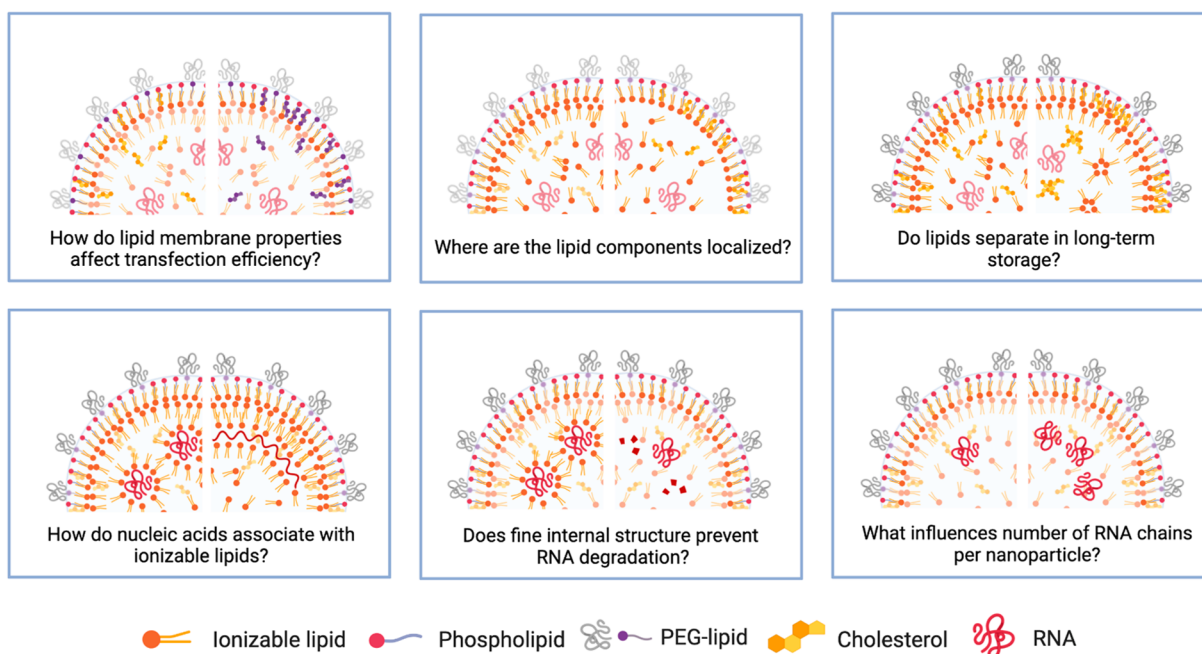


Figure 5. Unanswered questions remaining in the LNP structure. Figure created with BioRender.

which is part of the reason why intravenously administered LNPs typically end up in the liver. ApoE is intrinsically involved in cholesterol metabolism through forming a lipid complex and transporting it to the expressed low-density lipoprotein receptors (LDLRs), facilitating LDLR-mediated endocytosis. This concept was found to be applicable in the case of LNPs as well. Akinc et al. demonstrated that LNPs containing the ionizable lipid KC2 showed significantly lower cellular uptake in the absence of ApoE or LDLR compared with the normal conditions.⁴⁴ This observation was further supported by Dong et al., who reported that LNPs containing cKK-E12, an ionizable lipidoid, entered cells more efficiently in the presence of ApoE.⁷¹ Other lipids in LNP formulations also may be associated with ApoE binding to LNPs. Zhang et al. reported that DOPE induced greater interactions with ApoE than DSPC when the other lipid components in LNPs were identical, indicating that the changing the structure of the phospholipid alters the binding affinity of ApoE to LNPs.²⁰ Sebastiani et al. reported that adsorption of ApoE to mRNA-LNPs led to rearrangement of the lipid components in the nanoparticles, relocating a portion of cholesterol from the nanoparticle core to the shell upon ApoE exposure.³⁶ Kim et al. reported that modifications of the PEG content and conjugation of PEG-lipid with mannose allowed control over the cell-specific delivery of mRNA in the liver.⁷² Interestingly, ApoE is also abundant in the brain and central nervous system, where astrocytes produce it. Tanaka et al. explored delivering mRNA-LNPs via intracerebroventricular injection to mice and found that astrocytes successfully took up ssPalm LNPs.⁷³ However, ApoE is not the only known protein associated with differences in cellular uptake. Recently, Miao et al. demonstrated that serum albumin allows LNPs to enter cells via an ApoE-independent pathway,⁴³ suggesting the importance of other serum proteins. Despite these interesting observations, it is not yet clear how to fully harness the potential of the biomolecular corona. The biomolecular corona is responsible for LNP recognition by the immune system, blood circulation time, and biodistribution,⁷⁴ and finding ways to manipulate the

corona composition could lead to cell-, tissue-, or organ-specific uptake.

CONCLUSIONS AND FUTURE PERSPECTIVE

LNPs are a highly customizable nucleic acid delivery vector that has shown immense potential in mRNA vaccines. One should also not forget about their possible value in therapeutics for rare diseases and cancers. mRNA therapy can help produce therapeutic proteins to restore the functions of afflicted tissues or organs. An enormous amount of scientific research across the globe has been conducted to design and refine individual components of LNPs for efficient and safe delivery of the nucleic acid of interest. Nevertheless, it appears that the science of LNPs has just begun, and a number of unanswered questions remain. We have illustrated some of these questions in Figure 5. Undoubtedly, the surge of public interest in mRNA vaccines will expedite research efforts in this field, and we are carefully optimistic that we are witnessing the new era of nanomedicine.

AUTHOR INFORMATION

Corresponding Author

Gaurav Sahay – Department of Pharmaceutical Sciences, College of Pharmacy, Oregon State University, Portland, Oregon 97201, United States; Department of Biomedical Engineering, Oregon Health & Science University, Portland, Oregon 97201, United States; Department of Ophthalmology, Casey Eye Institute, Oregon Health & Science University, Portland, Oregon 97239, United States; orcid.org/0000-0003-1071-0500; Email: sahay@ohsu.edu

Authors

Yulia Eygeris – Department of Pharmaceutical Sciences, College of Pharmacy, Oregon State University, Portland, Oregon 97201, United States

Mohit Gupta – Department of Pharmaceutical Sciences,
College of Pharmacy, Oregon State University, Portland,
Oregon 97201, United States

Jeonghwan Kim – Department of Pharmaceutical Sciences,
College of Pharmacy, Oregon State University, Portland,
Oregon 97201, United States

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.accounts.1c00544>

Author Contributions

[†]Y.E. and M.G. contributed equally to the work.

Notes

The authors declare no competing financial interest.

Biographies

Yulia Eygeris is a postdoctoral scholar at Oregon State University with Prof. Gaurav Sahay. She attended Novosibirsk State University, where she earned a Specialist degree in Chemistry in 2014 under Dr. Igor Kirilyuk, and the University of Utah, where she completed her Ph.D. under Prof. Ilya Zharov in 2019. Her research interests include self-assembly and the chemistry of functional nanomaterials.

Mohit Gupta completed his Ph.D. in Medicinal Chemistry in 2019 at Duquesne University, where he worked on developing and optimizing novel molecules against HIV, breast cancer, and cognitive dysfunctions. As a postdoctoral scholar in Prof. Gaurav Sahay's lab at Oregon State University, he worked on developing novel materials for gene delivery applications. He recently joined GreenLight Biosciences Inc., where he is advancing the RNA delivery platform.

Jeonghwan Kim is a Ph.D. candidate in the laboratory of Prof. Gaurav Sahay at Oregon State University. He received his B.Pharm. and M.Sc. in Pharmacy from Yeungnam University in South Korea in 2012 and 2014, respectively. His research interests include the development of delivery platforms for RNA therapeutics and the intracellular trafficking of nanoparticles.

Gaurav Sahay received a B.S. in Pharmaceutical Studies from the University of Pune in 2003 and his Ph.D. at the University of Nebraska Medical Center under Prof. Alexander Kabanov in 2009. He then moved to Massachusetts Institute of Technology for postdoctoral work with Profs. Daniel Anderson and Robert Langer. In 2014 he joined the faculty of Oregon State University, where his research group now focuses on various aspects of mRNA delivery via lipid nanoparticles.

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